OM SHAH

Software Engineer

Tempe, AZ (Open to Relocate) | +1 201-266-3858| om.s@cvejobs.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience building scalable backend systems, distributed applications, cloud-native services, and AI-enabled enterprise platforms. Proficient in Python, Java, FastAPI, Spring Boot, AWS, Docker, and Kubernetes, with experience supporting production workloads exceeding 500K daily requests and 1M asynchronous events. Strong background in REST APIs, microservices, CI/CD, performance optimization, observability, infrastructure automation, and data processing, with experience integrating LLM, RAG, vector search, and intelligent retrieval capabilities into production software systems.

TECHNICAL SKILLS

Languages: Python, Java, C++, C#, JavaScript, SQL
Backend Engineering: FastAPI, Django, Spring Boot, .NET Core, REST APIs, gRPC, Microservices, API Design;
Distributed Systems & Messaging: Apache Kafka, Event-Driven Architecture, Asynchronous Processing, Distributed Systems
Cloud & DevOps: AWS (EC2, S3, Lambda, EKS, SQS), Docker, Kubernetes, Terraform, Helm, GitHub Actions, CI/CD
Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis, TimescaleDB, Pinecone, ChromaDB
Software Quality & Observability: Unit Testing, Integration Testing, Test Automation, Logging, Monitoring, Performance Tuning, Incident Triage
AI/ML: LLMs, Retrieval-Augmented Generation (RAG), LangChain, OpenAI API, Vector Databases, AI Agents, LangGraph, Embeddings, Semantic Search

PROFESSIONAL EXPERIENCE

Citigroup, AZ | Software Engineer — Nov 2025 – Present

- Designed and developed scalable backend systems and REST APIs using Python and Java for enterprise platforms, supporting 500K+ daily requests while improving platform reliability and request throughput.
- Built high-throughput microservices using FastAPI, Docker, and Kubernetes, reducing p95 API response latency by 28% under high-concurrency workloads.
- Designed scalable cloud-native services and distributed processing infrastructure supporting 1M+ asynchronous events per day while improving workload scalability and production reliability.
- Implemented CI/CD pipelines with automated testing, deployment automation, monitoring, telemetry, and logging for production applications, reducing average deployment time by 40%.
- Architected and integrated AI-powered services, including LLM and RAG capabilities, into five enterprise software platforms and business workflows.
- Built scalable retrieval, caching, and data-access systems supporting 75K+ indexed documents using Redis, vector search, and distributed data-processing technologies, reducing p95 retrieval latency by 30%.

Infinite Infolab, India | Software Engineer — Feb 2022 – Jul 2024

- Designed and developed scalable backend systems, REST APIs, and microservices in Python and Java, improving application performance, reliability, and p95 response time by 30%.
- Built asynchronous data and document-ingestion pipelines using FastAPI and AWS Lambda, processing up to 10K documents per day for scalable processing and workflow automation.
- Developed intelligent search, retrieval, and data-access capabilities across 500K+ indexed documents using Redis and vector databases, reducing average search latency by 25%.
- Integrated frontend and backend services to support real-time workflows, API-driven applications, and automated document processing, reducing average workflow completion time by 25%.
- Containerized and optimized microservices using Docker multi-stage builds, reducing container image size by 35% while improving deployment consistency, scalability, portability, and runtime efficiency.
- Collaborated with cross-functional teams to deploy secure cloud-native applications with authentication, monitoring, logging, observability, and scalable infrastructure, reducing production incident investigation time by 20%.

EDUCATION

Master of Science, Information Technology — **Aug 2024 – May 2026**
Arizona State University, USA
Bachelor of Engineering, Information Technology
Gujarat Technological University

SELECTED PROJECTS

AI-Powered Customer Support Platform | GitHub

- Built a production-style AI-powered customer support platform integrating LLM-based conversational workflows, real-time messaging, and Kafka-driven event streaming across distributed services.
- Designed scalable microservices with Redis caching and PostgreSQL-backed workflows supporting concurrent enterprise support operations with low-latency communication.
- Implemented JWT-secured authentication and containerized Kubernetes deployments, improving scalability, fault tolerance, and deployment consistency.

LLM-Powered Semantic Q&A Search Engine | GitHub

- Developed an LLM-powered semantic search engine leveraging vector embeddings and Retrieval-Augmented Generation (RAG) architecture for contextual enterprise question answering.
- Optimized inference and retrieval workflows to support 5K+ concurrent search requests under sub-500ms response latency across distributed backend services.
- Engineered scalable FastAPI-based APIs for intelligent document retrieval, contextual ranking, and AI-enhanced knowledge discovery, with MLflow-based tracking of model performance.

Cloud-Native Full Stack E-Commerce Platform | GitHub

- Built a scalable full-stack e-commerce platform using React, FastAPI, PostgreSQL, and AWS, supporting secure user authentication, product catalog management, and real-time order processing.
- Engineered microservices-based backend APIs with Redis caching and asynchronous processing, improving transaction handling performance and reducing API response latency.
- Implemented Dockerized Kubernetes deployments with CI/CD automation using GitHub Actions, enabling zero-downtime releases and reliable production monitoring.

CERTIFICATIONS

Machine Learning for Natural Language Processing | Python Programming